Radar HRRP Target Recognition Based on Dynamic Multi-task Hidden Markov Model

Lan Du, Penghui Wang, Hongwei Liu, Mian Pan and Zheng Bao
National Lab. of Radar Signal Processing, Xidian University, Xi'an, China
Email: dulan@mail.xidian.edu.cn

Abstract— **A Bayesian multi-task model is developed for radar automatic target recognition (RATR) using high-resolution range profile (HRRP). The aspect-dependent HRRP sequence is modeled using a stick-breaking hidden Markov model (SB-HMM) with time-evolving transition probabilities, in which the spatial structure across range cells is described by the hidden Markov structure and the temporal (or orientational) dependence between HRRP samples is described by the time evolution of the transition probabilities. This framework imposes the belief that temporally proximate HRRPs are more likely to be drawn from similar HMMs, while also allowing for possible distant repetition or "innovation" associated with abrupt fluctuation in the HRRP sequence. In addition, as formulated the stick-breaking prior and multi-task learning (MTL) mechanism are employed to infer the number of hidden states in an HMM and learn the target-dependent states collectively for all targets. The form of the proposed hierarchical model allows efficient variational Bayesian (VB) inference. The experimental results based on the measured HRRP data are compared with the MTL HMMs without time-evolution and also some other existing statistical models.**

I. Introduction

A high-resolution range profile (HRRP), a real vector, is composed of the amplitude of the coherent summations of the complex returns from target scatterers in each range cell. Since it contains abundant target information, such as target size, scatterer distribution, etc., target HRRP data can be used for radar automatic target recognition (RATR) [1]–[5], [7].

According to the scattering center model, HRRPs from complex targets yield target signatures that are a strong function of the target-sensor orientation, referred to as target-aspect sensitivity in [2], [3]. Nevertheless, there are generally contiguous sets of orientations (aspects) for which the scattering physics vary slowly, and for which the associated HRRPs can be approximated as statistically stationary. Therefore, we equally divided the HRRP frames according to the aspect sectors without most scatterers' motion through resolution cells(MTRC), and used distinct parametric models for statistical characterization of each HRRP frame in [2], [3], [7]. However, from the viewpoint of statistical recognition, the equal interval partition approach is not optimal.

To make use of the temporal dependence of multiple HRRPs from a sequence of target-sensor orientations, the hidden Markov model (HMM) has been utilized by [5] for radar HRRP sequence recognition, where each HMM "state" reflects a set of generally contiguous target-sensor orientations over which the HRRP statistics are relatively stationary, *i.e.* an aspect-frame we mentioned above, and the statistical relationships between a sequence of HRRPs are described via the transition probabilities from one state to the next under HMM formulation. Similar to the TFA model in [7], multiple aspect-dependent looks at a single target are utilized in the test phase. Since it is often difficult to achieve reliable classification based on a single view of the target, multi-aspect (sequence) recognition can be a better choice for RATR. Nevertheless, it also brings some challenges for the experimental system. Firstly, tracking radar is required for test phase to get an aspect-dependent sequence at a single target, which should be over several aspect-frames to reflect the aspect-dependent statistics in HMM; secondly, the angular sampling rates should remain same for both the training and test (classification) phases, since the temporal dependence is used as target's discriminative information. In fact, even if the second condition is satisfied, it is still hard to control the detailed temporal dependence between HRRP samples for moving targets, since the target pose is unknown and in-cooperative.

The research reported here seeks an alternative approach to characterize both the spatial structure across range cells and the temporal dependence between HRRP samples, in which the multi-aspect HRRP sequence is utilized for training model, but a single HRRP sample for test (classification) phase. In the proposed spatiotemporal model, the spatial structure across range cells is described by the hidden Markov structure and the temporal (or orientational) dependence between HRRP samples is described by the time evolution of the transition probabilities. As formulated the stick-breaking prior can effectively infer the optimal number of hidden states in an HMM, and the multi-task learning (MTL) mechanism, which learns the target-dependent states collectively for all targets, offers the potential to to improve overall recognition performance. In addition, the form of the proposed hierarchical model allows efficient variational Bayesian (VB) inference, of interest for the large-scale motivating RATR problem.

II. Bayesian Model Construction

For the HRRP target recognition problem, one typically has access to multi-aspect HRRP sequences from multiple targets. To model a multi-aspect HRRP sequence from target c ($c \in \{1, \cdots, C\}$) here, we divide the sequence into a series of subsequences $\{\boldsymbol{x}^{(c,m,n)}\}_{m=1,n=1}^{M_c,N}$, with $\boldsymbol{x}^{(c,m,n)} = [x^{(c,m,n)}(1), \cdots, x^{(c,m,n)}(T)]^{\mathrm{T}}$ representing the nth HRRP

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sample in the mth small set. Such a subsequence corresponds to a much smaller aspect-sector than that of the aspect-frame in [2], [3], [7], and the N samples can be assumed to be independent and identically distributed. In this paper we refer to these subsequences as sub-frames. Here we wish to learn a SB-HMM for each of these sub-frames to characterize the spatial structure across range cells, and further exploit the temporal dependence between sub-frames. If the learning of SB-HMM for data set $\{\boldsymbol{x}^{(c,m,n)}\}_{n=1}^{N}$ is termed learning task m from target c, the learning of SB-HMMs for all C targets, each with M_c tasks, jointly is referred to as multi-task learning (MTL).

To constitute a model with time-evolving state transition statistics, we consider a similar dynamic structure as in [6]. Specifically, the proposed model, termed the dynamic MTL-SB-HMM, is represented as

$$
\begin{aligned}
& x^{(c,m,n)}(t) \sim f(\theta_{s_t^{(c,m,n)}}) \\
& s_t^{(c,m,n)} \sim \begin{cases} \boldsymbol{w}_0^{(c,m)} & \text{if } t=1 \\ \boldsymbol{w}_{s_{t-1}^{(c,m,n)}}^{(c,m)} & \text{if } t>1 \end{cases} \\
& \boldsymbol{W}^{(c,m)} = [\boldsymbol{w}_0^{(c,m)\mathrm{T}}; \boldsymbol{w}_1^{(c,m)\mathrm{T}}; \cdots; \boldsymbol{w}_\infty^{(c,m)\mathrm{T}}] \\
& \boldsymbol{A}^{(c,m)} = [\boldsymbol{a}_0^{(c,m)\mathrm{T}}; \boldsymbol{a}_1^{(c,m)\mathrm{T}}; \cdots; \boldsymbol{a}_\infty^{(c,m)\mathrm{T}}] \\
& \begin{cases} \boldsymbol{W}^{(c,m)} = \boldsymbol{A}^{(c,m)} & \text{if } m=1 \\ \boldsymbol{W}^{(c,m)} = (1-\rho_m^{(c)})\boldsymbol{W}^{(c,m-1)} + \rho_m^{(c)}\boldsymbol{A}^{(c,m)} & \text{if } m>1 \end{cases} \\
& a_{i,j}^{(c,m)} = v_{i,j}\prod_{h=1}^{j-1}(1-v_{i,h}) \ , \quad v_{i,j} \sim \mathrm{Beta}(1,\alpha_i) \\
& \alpha_i \sim \mathrm{Ga}(a_\alpha, b_\alpha) \ , \quad \rho_m^{(c)} \sim \mathrm{Beta}(1, f_0) \ , \quad \theta_i \sim H \\
& \qquad j = 1, \cdots, \infty \ ; \quad i = 1, \cdots, \infty
\end{aligned} \tag{1}
$$

where $x^{(c,m,n)}(t)$ denotes the nth sample's amplitude echo in the tth range cell in the mth sub-frame of the cth target; $s_t^{(c,m,n)}$ is the corresponding state indicator; $\boldsymbol{W}^{(c,m)}$ is the corresponding infinite state transition matrix; each row of the corresponding innovation state-transition matrix $\boldsymbol{A}^{(c,m)}$ is given a stick-breaking prior a_α, b_α and f_0 are preset hyperparameters. In this model, we learn SB-HMMs with aspect-dependent state transition matrices for all of the tasks (sub-frames) from a given target as well as share the same state set across all tasks.

To impose the time-evolution across all sub-frames form a given target, $\boldsymbol{W}^{(c,m)}$ is modified from $\boldsymbol{W}^{(c,m-1)}$ by introducing a new innovation state-transition matrix $\boldsymbol{A}^{(c,m)}$, and the random variable $\rho_m^{(c)} \in [0,1]$ controls the degree of innovation in $\boldsymbol{W}^{(c,m)}$ relative to $\boldsymbol{W}^{(c,m-1)}$. The dynamic structure in (1) can further be extended

$$
\begin{aligned}
\boldsymbol{W}^{(c,m)} &= (1-\rho_m^{(c)})\boldsymbol{W}^{(c,m-1)} + \rho_m^{(c)}\boldsymbol{A}^{(c,m)} \\
&= \prod_{k=1}^{m-1}(1-\rho_k^{(c)})\boldsymbol{W}^{(c,1)} + \sum_{k=1}^{m-1}\left\{\prod_{h=k+1}^{m-1}(1-\rho_h^{(c)})\right\}\rho_k^{(c)}\boldsymbol{A}^{(c,m)} \\
&= u_{m1}^{(c)}\boldsymbol{A}^{(c,1)} + u_{m2}^{(c)}\boldsymbol{A}^{(c,2)} + \cdots + u_{mm}^{(c)}\boldsymbol{A}^{(c,m)}
\end{aligned} \tag{2}
$$

where $u_{mk}^{(c)} = \rho_{k-1}^{(c)}\prod_{h=k}^{m-1}(1-\rho_h^{(c)})$, for $k = 1, \cdots, m$, with $\rho_0^{(c)} = 1$. It can easily verify that $\sum_{k=1}^{m} u_{mk}^{(c)} = 1$ for each m, with $u_{mk}^{(c)}$ denoting the prior probability that state-transition matrix $\boldsymbol{W}^{(c,m)}$ is drawn from the $\{\boldsymbol{A}^{(c,1)}, \boldsymbol{A}^{(c,2)}, \cdots, \boldsymbol{A}^{(c,m)}\}$. The global HMM and independent MTL-HMM across all tasks (sub-frames) from each target are the limiting cases of this model:

- when $\rho_m^{(c)} \to 0$, $\boldsymbol{W}^{(c,m)} \to \boldsymbol{W}^{(c,m-1)}$ and there is no innovation, resulting in a common state-transition matrix for all sub-frames from target c;
- when $\rho_m^{(c)} \to 1$, $\boldsymbol{W}^{(c,m)} \to \boldsymbol{A}^{(c,m)}$, where the new innovation $\boldsymbol{A}^{(c,m)}$ controls the time-evolution, resulting in each sub-frame from target c having a unique state-transition matrix.

The form of the observation model $f(\cdot)$ may vary depending on the application; in the work presented here, since the normal distribution are used to describe HRRP data in [2], [4], [7] with the good performance, $f(\cdot)$ is defined as a univariate Normal distribution, and $H(\cdot)$ is a Normal-Gamma distribution to preserve the conjugacy requirements.

III. Experimental Results

We examine the performance of the proposed model on the measured airplane data used in [2], [3], [7]. The center frequency and bandwidth of the radar are 5520MHz and 400MHz. The projections of target trajectories onto the ground plane are shown in Figure 1. In order to test the generalization performance of the recognition methods, the 2nd and the 5th segments of Yak-42, the 6th and the 7th segments of Cessna, the 5th and the 6th segments of An-26 are taken as the training samples, other data segments are taken as test samples in our experiments. With large SNR, the effect of noise in the measured data on the recognition can be ignored. In addition, the measured HRRP is a 256-dimensional vector.

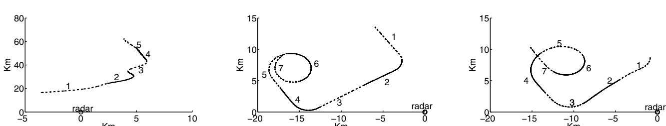

Fig. 1. Projections of target trajectories onto the ground plane: (a) Yark-42; (b) Cessna Citation S/II; (c) An-26.

As discussed in Section II, here the whole HRRP sequence from each target is first equally segmented into some sub-frames, then our algorithm can adaptively combine some of them and infer the posterior frame segmentation. It is important to restate that these initial sub-frames should be much smaller than those used in [2], [3], [7]. In our previous experiments in [2], [3], [7], there are totally 135 aspect-frames, of which each frame has 1024 HRRP samples; while in the following experiments, each sub-frame has 256 HRRP samples, and there are totally 540 sub-frames.

The recognition accuracies of the dynamic MTL-SB-HMM and the independent MTL-SB-HMM versus the size of training dataset are shown in Figure 2, of which the latter one is the limiting case of our proposed model with $\rho_m^{(c)} \to 1$ for $m =$

$1, \cdots, M_c$ and $c = 1, 2, 3$, *i.e.* the time-evolving character is removed. Here we utilized the six training datasets: 1×540, 2×540, 4×540, 8×540, 16×540 and 32×540, where 8×540 means that 8 HRRP samples randomly selected from each of the 540 sub-frames, similarly for other data size expressions. The recognition rates in Figure 2 are averaged over five experiments, and the error bars denote the corresponding standard deviations. From Figure 2, we can clearly see that the dynamic MTL model has better recognition performance than the independent MTL model. In addition, the larger training dataset generally yields the higher and stabler recognition rate.

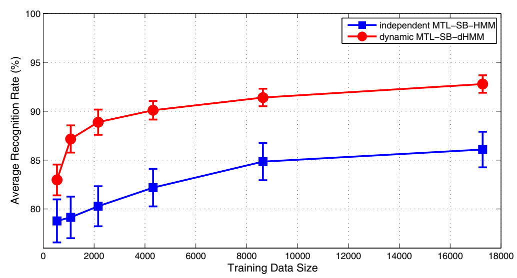

Fig. 2. Recognition accuracies of the dynamic MTL-SB-HMM and the independent MTL-SB-HMM versus the size of training dataset. The recognition rates are averaged over five experiments, and the error bars denote the corresponding standard deviations.

For aircraft-like targets, the noises in the inphase and quadrature echoes of targets can be assumed to be Gaussian white noises. In order to evaluate the effect of noises on the proposed model and other existing statistical models, we add simulated white noises to the the inphase and quadrature component of the raw test data. Figure 3 depicts the average recognition rates versus SNR, generated via the five statistical models, Gaussian model [4], Gamma model [1], the two-distribution compounded model [3], FA model [2] and the new dynamic MTL model. Obviously, the Gaussian model with power transformation is most sensitive to noises. When $\mathrm{SNR} \geq 35\mathrm{dB}$, the FA model with power transformation outperforms other four models; when $\mathrm{SNR} \in [30\mathrm{dB}, 20\mathrm{dB}]$, our two-distribution model achieves the best performance; when $\mathrm{SNR} \leq 15\mathrm{dB}$, only the FA model and the dynamic MTL model can yield recognition rates larger than 60%. Although the comparable recognition accuracies got by the two models (especially with the low SNR), it is important to point out that the FA model requires much more training data than our new model, due to the full covariance estimation for the 256-dimensional HRRP sample.

Table I summarizes the corresponding model selection results for the different training data. Generally, the number of meaning states is less than 25 and that of segmented frames less than 45. Please note the inferred number of frames is much less than 135, which is the number of frames used for other models [2], [3], [7], therefore, the computation burden in the test (classification) phase can be reduced.

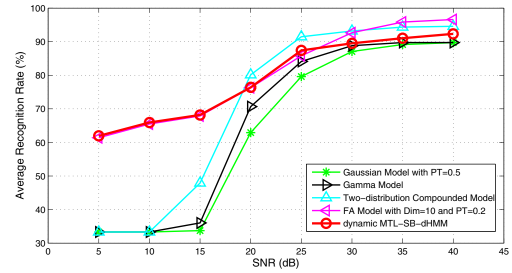

Fig. 3. Variation of the recognition performance with SNR, via Gaussian model with power transformation ($\alpha = 0.5$) [4], Gamma model [1], our two-distribution compounded statistical model [3], FA model with the dimensionality of the latent factor variable $D = 10$ and power transformation ($\alpha = 0.2$) [2], and the proposed dynamic MTL-SB-HMM. Here the dynamic MTL model is trained with $32 \times 540 = 17280$ training data; while the other four models with $1024 \times 135 = 138240$ training data. The recognition rates are averaged over five experiments.

TABLE I

Numbers of meaningful states and segmented frames for different training HRRP data, inferred via the dynamic MTL-SB-HMM. Here we give the average results over five experiments for each training data size.

data $\sharp$ per sub-frame	1	2	4	8	16	32
average state $\sharp$	18.4	21.6	22.4	22.8	21.6	23.2
average frame $\sharp$	23.6	23.6	29.2	30.6	35.4	40.6

IV. Conclusion

We have developed a dynamic Bayesian spatiotemporal multi-task learning model for radar HRRP data, termed the dynamic multi-task learning based hidden Markov model with stick-breaking prior (dynamic MTL-SB-HMM). The proposed construction allows the HMM states shared for all tasks (here the multiple tasks are linked to specific aspect-frames) and the convenient VB inference, of interest for the large-scale motivating RATR problem. The algorithm has been demonstrated on measured HRRP data.

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